Introduction to Software Engineering

COSC 336

ReSource (Phase III)

Requirements Document

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Intelligent AI-Powered Circular Campus Resource Exchange and Asset Life Cycle Management System

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This file contains the parts of the ReSource Phase 3 Requirements Document prepared by Mohammad Alsuwaidi (100064887). Numbering, requirement IDs and cross-references are the same as in the combined group document.

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2. System Features

2.1 Asset Registration (FR-1)

2.1.1 Description and Priority

This feature lets an Asset Coordinator record unused, surplus or reusable assets with enough detail for other departments to find, judge and request them. Every later step, including search, matching, transfer, maintenance and end-of-life decisions, depends on these records being complete and correct.

Priority: High. This feature develops Phase 2 requirement FR-1. Asset registration and status management are part of the minimum prototype [1], and the other features depend on the asset records it creates.

2.1.2 Stimulus/Response Sequences

1. The Asset Coordinator chooses to register an asset. The system opens a new record for the Asset Coordinator's department.

2. The Asset Coordinator enters the asset details, including the hazard flag, and adds photographs and supporting documents. The system checks each value and file.

3. The Asset Coordinator may ask for AI suggestions (UC-10). The system shows them separately, and nothing is saved until the Asset Coordinator accepts it.

4. The Asset Coordinator saves a draft or publishes. The system checks the required fields, gives the asset a unique ID and sets its state to Draft or Available.

5. If a field is missing, a value is invalid or a file is refused, the system names the problem and keeps everything else that was entered.

2.1.3 Functional Requirements

ID	Requirement	Priority	Use case	Acceptance criterion (done when…)
FR-1.1	The system shall allow only Asset Coordinators to create asset records, and only for the departments they are assigned to.	Essential	UC-1	An Asset Coordinator can save a record for their own department. A Department Representative, or an Asset Coordinator of another department, is refused and nothing is saved.
FR-1.2	The system shall require these fields before an asset is published: name, description, category, quantity, Asset Coordinator, owning department, location (building and room), condition and hazard flag (whether the asset is hazardous or electronic waste).	Essential	UC-1	Publishing with any of these fields empty is blocked, and each missing field is named.

ID	Requirement	Priority	Use case	Acceptance criterion (done when…)
FR-1.3	The system shall allow these optional fields: subcategory, specifications, brand and model, serial or tag number, purchase date, first-allocation date, estimated value (AED), replacement price (AED), weight (kg), material type, and handling or safety notes (for example heavy, fragile or contains batteries).	Essential	UC-1	Optional fields can be saved, viewed and edited. A record without them can still be published.
FR-1.4	The system shall record condition only from the list New, Good, Fair or Poor. Availability shall be set by the system from the asset state (Appendix B.1) and shall not be typed by users.	Essential	UC-1	Condition accepts only the listed values. After every change, the shown availability matches the asset state.
FR-1.5	The system shall reject invalid values: quantity must be a whole number of at least 1, estimated value, replacement price and weight cannot be negative, purchase and first-allocation dates cannot be in the future, and category must come from the approved list.	Essential	UC-1	Test entries of quantity 0, value -50, weight -1, a future purchase date and an unknown category are each rejected with a message. Other entered data is kept.
FR-1.6	The system shall allow up to 5 photographs and up to 3 supporting documents (for example invoice, manual or warranty) per asset. Each file shall be an image or a PDF of at most 10 MB (SEC-13, TBD-13).	Essential	UC-1	Five photographs and three documents attach. A sixth photograph, a file over 10 MB or a file that is not an image or PDF is refused and the limit is shown.
FR-1.7	The system shall give each asset a unique asset ID that is never reused, even after the asset is archived.	Essential	UC-1	100 test records receive 100 different IDs. Archiving a record does not free its ID.
FR-1.8	The system shall warn when a serial or tag number already exists on another record.	Essential	UC-1	Entering an existing serial number shows a warning that links to the other record. The new record saves only after the Asset Coordinator corrects it or confirms it is a different item.

ID	Requirement	Priority	Use case	Acceptance criterion (done when…)
FR-1.9	The system shall let the Asset Coordinator save an asset as a draft and publish it later. Drafts shall not be visible to other departments.	Essential	UC-1, UC-2	A draft does not appear in search. After publishing, the asset appears in search with the state Available.
FR-1.10	The system shall not allow changes to category, quantity or condition while the asset is Reserved or In transfer.	Essential	UC-1, UC-5	Editing these fields on a Reserved asset is refused with the reason. The same edit on an Available asset is saved.
FR-1.11	The system shall let the Asset Coordinator withdraw a published asset from listing when it has no open request.	Desirable	UC-1	A withdrawn asset no longer appears in search. Withdrawal is refused while a request for the asset is still open.
FR-1.12	The system shall allow registration to be completed without AI. AI-suggested values shall be saved only after the Asset Coordinator accepts them (see FR-9).	Essential	UC-1, UC-10	With AI unavailable, a full registration still completes. No AI value appears in a saved record unless the Asset Coordinator accepted it.
FR-1.13	The system shall record the creation of each asset and every later change, with the user, time, and old and new values.	Essential	UC-1	Every create, edit, publish and withdrawal appears in the audit log (FR-7.A) with user, time and changed fields.
FR-1.14	The system shall allow many assets to be registered at once from one uploaded list.	Future	UC-1	A list of 20 assets with 2 invalid rows creates 18 drafts and reports the 2 rejected rows with reasons.
FR-1.15	The system shall not publish an asset flagged as hazardous or electronic waste until a Sustainability Officer has recorded a safety review that allows it to be offered for reuse (SAF-1).	Essential	UC-1	Publishing a hazardous test asset without a recorded safety review is refused with the reason. After the review is recorded, publication succeeds.

2.3 Resource Requests and Reservations (FR-3)

2.3.1 Description and Priority

This feature lets a Department Representative ask for a specific published asset, or describe a need when nothing suitable is listed. Submitting a request does not reserve the asset; the requested quantity is reserved only when the request is approved (FR-4), so the same units are never promised to two departments. The Department Representative can follow the request until the asset is received.

Priority: High. This feature develops Phase 2 requirement FR-3. Resource-request submission and tracking are part of the minimum prototype [1].

2.3.2 Stimulus/Response Sequences

1. The Department Representative opens a published asset and chooses to request it, or creates a request for a need. The system fills in any details it already knows.

2. The Department Representative enters the purpose, quantity, required date and other request details. The system checks each value.

3. The Department Representative submits. The system checks the current available quantity, gives the request an ID, sets it to Submitted and notifies the owning Asset Coordinator. No units are reserved at this point.

4. For a need with no chosen asset, the system sets the request to Open so the Procurement Team can look for matches (UC-9).

5. The Department Representative follows the state of the request. They can edit it while it is Submitted or cancel it before handover; cancelling an approved request releases its reservation.

6. If a value is invalid, the quantity is no longer available or the asset is restricted, the system explains the problem and keeps the entered details.

2.3.3 Functional Requirements

ID	Requirement	Priority	Use case	Acceptance criterion (done when…)
FR-3.1	The system shall allow a Department Representative to request a specific published asset for a quantity no greater than its current available quantity.	Essential	UC-3	A request for 2 of 5 available units is saved. A request for 6 of 5 is refused and the available quantity is shown.
FR-3.2	The system shall allow a Department Representative to submit a request for a need when no listed asset fits, without choosing an asset.	Essential	UC-3, UC-9	The request is saved as Open and appears in the Procurement Team’s list for matching.
FR-3.3	The system shall require: requesting department, category, purpose, quantity, required date and delivery location. Specifications, preferred minimum condition and urgency (Low / Medium / High, default Medium) shall be optional.	Essential	UC-3	Submitting with any required field empty is blocked and each missing field is named.
FR-3.4	The system shall reject a quantity below 1, a required date in the past, and a preferred condition or category not on the approved lists.	Essential	UC-3	Each invalid test value is rejected with a message, and the other entered details are kept.

ID	Requirement	Priority	Use case	Acceptance criterion (done when…)
FR-3.5	The system shall allow a Department Representative to submit requests only for their own department.	Essential	UC-3	Choosing another department is refused. A Department Representative can see every request from their department and no others.
FR-3.6	The system shall give each submitted request a unique request ID, set its state to Submitted and notify the owning Asset Coordinator.	Essential	UC-3	Each new request shows its own ID and the state Submitted, and the owning Asset Coordinator receives a notification (FR-7.N1).
FR-3.7	The system shall check the available quantity again at the moment of submission and refuse a request for more units than are currently available. Submitting a request shall not reserve units or change the asset's ownership, custody or location.	Essential	UC-3	With 8 units available, a request for 5 is saved and the available quantity stays 8. A request for 9 is refused and the current available quantity is shown.
FR-3.8	The system shall reserve the requested quantity only when the request receives its final required approval (FR-4.5), and shall release the reservation when an approved request is cancelled or expires. Two approved requests shall never reserve the same units.	Essential	UC-3, UC-4	With 8 units available, approving a request for 5 leaves 3 available. When two requests for the last unit are approved at the same moment, exactly one succeeds and the available quantity never goes below zero. Cancelling the approved request restores the quantity.
FR-3.9	The system shall expire a request that is not decided within 10 working days of submission, release the reservation of an approved request that is not handed over within 10 working days of approval, and notify both departments (TBD-5).	Desirable	UC-3, UC-4, UC-5	A test request left undecided for 10 working days becomes Expired. An approved test request not handed over in 10 working days becomes Expired, its quantity is released, and both departments are notified.
FR-3.10	The system shall show the Department Representative the current state of each request and its full history, with date, person and reason for every change.	Essential	UC-3	The request page lists every state change in order, with date, person and reason.

ID	Requirement	Priority	Use case	Acceptance criterion (done when…)
FR-3.11	The system shall allow the requesting Department Representative to edit a request while it is Submitted, and to cancel it at any time before handover.	Essential	UC-3	Edits save on a Submitted request and are refused once it is Released. Cancelling an Approved request sets it to Cancelled and releases its reservation.
FR-3.12	The system shall not allow the requesting Department Representative to release, approve or change the state of their own request in any way other than editing or cancelling it.	Essential	UC-3, UC-4	A test attempt by the requesting Department Representative to release or approve their own request is refused and recorded in the audit log.
FR-3.13	The system shall warn when the same department already has an open request for the same asset.	Desirable	UC-3	A second request from the same department for the same asset shows a warning with a link to the first request.
FR-3.14	Requests created through the AI assistant shall pass the same checks and need the same confirmation as requests made on the form (see FR-11.2 and FR-11.3).	Desirable	UC-3, UC-11	A request made through the assistant has the same fields, checks and state as one made on the form, and is not submitted until the Department Representative confirms.
FR-3.15	The system shall refuse a request for an asset that has an open maintenance request, a pending end-of-life proposal, an unsafe status or an unreviewed hazard flag (FR-5.9, FR-5.27, SAF-1, SAF-5), and shall name the restriction.	Essential	UC-3	A request for each type of restricted test asset is refused and the message names the restriction. The same request for an unrestricted asset is saved.

2.4 Approvals and Transfers (FR-4)

2.4.1 Description and Priority

This feature moves an asset between departments in four controlled steps: release by the owning department's Asset Coordinator, approval, transfer by collection or delivery, and receipt. The requested quantity is reserved at final approval, and custody and location change only when the receiving department confirms receipt. Every decision is made by an authorized person; AI may recommend but never decides.

Priority: High. This feature develops Phase 2 requirement FR-4. Transfer approval and confirmation and ownership tracking are part of the minimum prototype [1].

2.4.2 Stimulus/Response Sequences

1. The Asset Coordinator of the owning department reviews a new request and releases it or declines it with a reason. The system sets the request to Released or Rejected.

2. The Administrator approves or rejects the released request. Requests above the approved value limit, or of unknown value, also need a Procurement Team member's approval (FR-4.3). On final approval, the system rechecks availability and reserves the requested quantity.

3. The Asset Coordinator arranges the transfer: collection or delivery, planned date, from and to locations, contact person and any transfer cost. The system creates a Scheduled transfer and notifies the requesting Department Representative.

4. At handover, the Asset Coordinator marks the asset as handed over. The system checks the safety conditions and sets the transfer and asset to In transfer.

5. A Department Representative or the Asset Coordinator of the receiving department confirms receipt and records the condition on arrival. The system changes the Asset Coordinator, department and location, and completes the request.

6. If the asset arrives damaged or is the wrong item, the receiver reports a problem. The system sets the transfer to Disputed and asks the Administrator to decide.

2.4.3 Functional Requirements

ID	Requirement	Priority	Use case	Acceptance criterion (done when…)
FR-4.1	The system shall ask the owning department's Asset Coordinator to release or decline each new request. Declining shall need a reason.	Essential	UC-4	A request cannot reach approval until it is released. A decline without a reason is refused.
FR-4.2	The system shall send released requests to the Administrator to approve or reject. Rejecting shall need a reason.	Essential	UC-4	Approve and reject both work. Every rejection shows its reason to the requesting Department Representative.
FR-4.3	The system shall require a second approval from a Procurement Team member when the estimated value is above a limit approved by the university and configured by the System Administrator (TBD-2), or when the value is unknown.	Desirable	UC-4	A test request above the limit cannot become Approved until both approvals are recorded.
FR-4.4	The system shall not allow one person to approve a request they submitted, or to both release and approve the same request.	Essential	UC-4	Both test attempts are refused and recorded in the audit log.

ID	Requirement	Priority	Use case	Acceptance criterion (done when…)
FR-4.5	The system shall recheck availability, quantity, safety restrictions and permissions just before the final approval is saved, and shall reserve the requested quantity in the same step (FR-3.8).	Essential	UC-4	If the asset was withdrawn, became restricted or now has fewer available units than requested, approval is blocked and the Administrator sees the up-to-date details. A successful approval reduces the available quantity by the requested amount.
FR-4.6	The system shall notify the requesting Department Representative and the owning Asset Coordinator of every release, approval, rejection and transfer step.	Essential	UC-4, UC-5	In testing, both parties receive a notification (FR-7.N) within 5 minutes of each decision. If sending fails, the decision is still saved.
FR-4.7	The system shall let the owning Asset Coordinator create a transfer only for an approved request, with method (collection or delivery), planned date, from location, to location, contact person and, optionally, the transfer cost in AED.	Essential	UC-5	A transfer cannot be created for a request that is not Approved. Missing transfer fields are named, and a negative transfer cost is refused.
FR-4.8	The system shall allow supporting documents, such as a handover form, delivery note or photographs at handover, to be attached to a transfer (up to 5 files).	Desirable	UC-5	Five files attach to a transfer and can be viewed by both departments. A sixth is refused.
FR-4.9	The system shall set the asset and transfer to In transfer when the Asset Coordinator marks the asset as handed over. Before handover, the Asset Coordinator may reschedule or cancel; cancelling returns the reserved units to Available.	Essential	UC-5	Handover sets both states to In transfer. Cancelling a Scheduled transfer restores the available quantity.
FR-4.10	The system shall allow only a Department Representative or the Asset Coordinator of the receiving department to confirm receipt, and shall require the condition on arrival.	Essential	UC-5	A user from another department cannot confirm receipt. Confirmation without a condition is refused.

ID	Requirement	Priority	Use case	Acceptance criterion (done when…)
FR-4.11	On confirmed receipt, the system shall change the asset's Asset Coordinator, owning department and location, and set the request to Completed. These details shall not change before receipt.	Essential	UC-5	After receipt, the asset shows the new Asset Coordinator, department and location. Before receipt, they are unchanged.
FR-4.12	The system shall let the receiver report a problem on arrival. The transfer shall become Disputed and the Administrator shall decide to accept it or return the asset.	Essential	UC-5	A problem report sets the transfer to Disputed and notifies the Administrator. Custody does not change until the dispute is decided.
FR-4.13	The system shall remind the receiving department if receipt is not confirmed within 5 working days of handover, with a copy to the Administrator.	Desirable	UC-5	A test transfer left unconfirmed for 5 working days sends one reminder to the receiver and the Administrator.
FR-4.14	The system shall keep the full ownership and transfer history of every asset and record every release, approval, transfer and receipt step.	Essential	UC-4, UC-5	The asset page lists every past owner with dates. Each step appears in the audit log (FR-7.A) with user, time and decision.
FR-4.15	When only part of a multi-unit asset is transferred, the system shall create a linked record for the transferred units and reduce the original quantity.	Essential	UC-5	Transferring 2 of 5 chairs leaves 3 on the original record and creates a new record of 2, linked to the original, for the receiving department.
FR-4.16	The system shall block handover of an asset that is unsafe, under maintenance or pending an end-of-life action, and of a risky asset without a recent safe inspection (SAF-5, SAF-9, FR-5.9, FR-5.27). It shall show any handling or safety notes to the receiving department before handover (SAF-7).	Essential	UC-5	Handover of each type of blocked test asset is refused with the reason. For a test asset with safety notes, the notes appear on the receiving department's transfer screen.

4. Other Requirements

4.1 Data Requirements

The system shall store the following information for each entity. IDs are never reused, records involved in a request or transfer are archived rather than deleted, money values are in UAE dirhams (AED), weights are in kilograms, and dates and times use UAE time. Retention periods follow Section 4.2 and the university records policy once confirmed (TBD-4). Personal data is limited to the fields in SEC-7.

Entity	Field	Description	Required?
User	User ID	Unique internal ID for each account; used in audit entries instead of personal data	Yes
User	Name; university ID; university email	Identify the user; the email is used for sign-in and notifications	Yes
User	Password	Stored only in a protected, non-reversible form (SEC-6)	Yes
User	Role(s); department	One or more roles from Section 5.1 and the department the user belongs to	Yes
User	Account status	Pending approval / Active / Suspended / Closed (Appendix B.6)	Yes
User	Consent record	Consent given or withdrawn, date and time, and version of the privacy notice and terms (FR-6.2, SEC-8)	Yes
User	Failed sign-in count; locked until	Used for account lockout (SEC-3)	Yes
Department	Department ID; name; status	Unique code, full name, and Active / Inactive (FR-6.9)	Yes
Department	Department Representative	User who represents the department	Yes
Category	Category ID; name; parent category; status	Entry in the approved category list; parent for a subcategory; Active / Retired	ID and name: Yes; others: No
Category default value	Category; value type; value; unit; source; effective date; version	Replacement price (AED), storage and handling cost (AED), default weight (kg), additional years of use and emission factor (kg CO_2e per item), kept in every version (FR-13.4, FR-13.5)	Yes

Entity	Field	Description	Required?
Asset	Asset ID	Unique ID, never reused	Yes
Asset	Name; description	Short name and full description	Yes
Asset	Category; subcategory	From the approved list	Category: Yes; subcategory: No
Asset	Quantity; available quantity	Total units, and units not reserved or otherwise restricted	Yes
Asset	Asset Coordinator; owning department	Person and department responsible for the asset	Yes
Asset	Location	Building and room	Yes
Asset	Condition	New / Good / Fair / Poor	Yes
Asset	Hazard flag; safety review	Whether the asset is hazardous or electronic waste, and the recorded safety review (FR-1.15, FR-5.23)	Hazard flag: Yes; review: when required
Asset	Safe-to-use status; date of last inspection	Result of the latest inspection (SAF-5, SAF-9)	When inspected
Asset	State	One state from Appendix B.1, set by the system	Yes
Asset	Purchase date; first-allocation date; estimated value (AED); replacement price (AED); weight (kg)	Used for age, time in service and the indicators (FR-5.19, FR-13.6)	No
Asset	Specifications; brand and model; serial or tag number; material type; handling or safety notes	Technical, identifying and handling details	No
Asset	Photographs; supporting documents	Up to 5 photographs and 3 documents, images or PDF of at most 10 MB each	No
Asset	Cumulative maintenance cost (AED); end-of-life review flag	Calculated by the system (FR-5.10, FR-5.11)	Yes
Asset	Parent asset ID	Link from a split or component record to the original asset (FR-4.15, FR-5.33)	No
Asset	Created by / updated by; dates	Who created and last changed the record, and when	Yes

Entity	Field	Description	Required?
Request	Request ID; requesting Department Representative; department	Unique ID, person and department asking	Yes
Request	Asset ID	Requested asset; empty for an Open need until a match is chosen	No
Request	Category; purpose; quantity; required date; delivery location	What is needed, why, how many, by when and where	Yes
Request	Specifications; preferred condition; urgency	Extra detail; minimum condition; Low / Medium / High	No
Request	State; history	One state from Appendix B.2, with every change, person and reason	Yes
Match	Match ID; request ID; asset ID	Link between a request and a suggested asset	Yes
Match	Score; confidence; reasons	Score 0 to 100, confidence level (AI-10) and explanation (FR-8.4)	Yes
Match	Rule or model version; reviewer decision and reason	Version used and the Procurement Team member's choice and reason	Yes
Reservation	Reservation ID; request ID; asset ID; quantity reserved	Units reserved when a request receives its final approval (FR-3.8)	Yes
Reservation	Start; expiry; status	When the reservation began and when it expires; Active / Released / Expired / Completed	Yes
Approval	Approval ID; request ID; step	Release / Approval / Second approval	Yes
Approval	Decision; reason; decided by; date and time	Who decided what, when and why	Yes (reason required for a decline or rejection)
Transfer	Transfer ID; request ID; asset ID; quantity	Units being moved for a request	Yes
Transfer	Method; from location; to location; planned date; contact person; transfer cost (AED)	Collection or delivery, where to, and the cost used in FR-13	Yes (transfer cost: No)
Transfer	Handover date; receipt date; received by; condition on arrival	Recorded at handover and receipt	Yes, when each step happens

Entity	Field	Description	Required?
Transfer	Supporting documents; dispute notes; state	Up to 5 files; problem details; state from Appendix B.3	Documents: No; state: Yes
Maintenance request	Request ID; asset ID; reported by; problem; assigned staff; state	Defect report or inspection request; state from Appendix B.4	Yes
Maintenance request	Inspection findings; safe to use; work type; work performed; parts; labor hours; cost (AED); additional years of use; start and completion dates	Inspection and repair results (FR-5.5, FR-5.6)	When the work is recorded
End-of-life record	Record ID; asset ID; action; proposer; justification; hazard confirmation; safety reviewer; handling party; approver; state	Proposal, safety review and approval (FR-5.20 to FR-5.28); state from Appendix B.5	Yes
End-of-life record	Recipient or handling party; date; quantity; weight (kg) and weight type; handover document or certificate; final location	Completion details (FR-5.29 to FR-5.32)	Yes (certificate required for hazardous and electronic-waste items)
Approved vendor	Vendor ID; name; materials handled; contact details; approval expiry date	Approved recycling and disposal vendors (FR-5.25)	Yes
Life-cycle event	Event ID; asset ID; event type; date and time; performed by; approved by; related record	Permanent asset history; events cannot be edited or deleted (FR-5.16, FR-5.18)	Yes
Sustainability impact	Period; scope; indicator; value; unit; share of default values; category value versions used	Calculated sustainability indicators (FR-13)	Yes
Notification	Notification ID; recipient; event; linked record; created time; read status; delivery status; retry count	Messages sent to users (FR-7.N)	Yes
Audit log entry	Entry ID; user ID; role; action; record type and ID; old and new values; date and time	Permanent record of every action; entries cannot be edited or deleted (FR-7.A)	Yes

Entity	Field	Description	Required?
AI output record	Output ID; AI feature; time; input record references; model and rule version; result; confidence; explanation; human decision and reason	Trace of every AI result used in a decision (AI-5, FR-7.A4)	Yes
Human review request	Review ID; AI output ID; requested by; reason; Administrator decision; dates	Requests for human review of AI results (SEC-10)	Yes
AI report	Report ID; type; period; scope; draft text; approved text; model version; data sources; edited by; approved by; state	AI-generated reports; state from Appendix B.7 (FR-12.10)	Yes
Model version	Version ID; AI component; version label; in use from / to; approved by	Which AI version produced each suggestion	Yes

5. Use Cases

UC-1: Register and Publish an Asset

Use case ID and name	UC-1: Register and Publish an Asset
Description	The Asset Coordinator records an unused or reusable asset and publishes it so other departments can find and request it.
Primary actor (initiator)	Asset Coordinator
Secondary actors	None. UC-10 may extend this use case when the Asset Coordinator asks for AI suggestions.
Related requirements	FR-1.1 to FR-1.15; FR-2.1; FR-5.16, FR-5.33; FR-13.6; PER-3, PER-6; SAF-1, SAF-5; SEC-13; QUA-1, QUA-4, QUA-7; BR-2.
Preconditions	The Asset Coordinator is signed in and assigned to the department that holds the asset. The approved category list is available.
Trigger	The Asset Coordinator chooses to register an asset.
Main flow	1. The Asset Coordinator opens a new asset record for their department. 2. The Asset Coordinator enters the required and optional details, including the hazard flag, and attaches photographs or documents. 3. The system checks every value and file and shows the completed record. 4. The Asset Coordinator chooses to publish. 5. The system gives the asset a unique ID, sets its state to Available and shows it in search. 6. The system records the action in the audit log.
Alternative flows	A1 at step 4: The Asset Coordinator saves a draft. The state is Draft and the asset is not visible to other departments. A2 at step 2: The Asset Coordinator asks for AI suggestions (UC-10) and accepts, edits or rejects them. Continue at step 3. A3 at step 5: The asset is flagged as hazardous. The system keeps it as Draft until a Sustainability Officer records a safety review (FR-1.15, SAF-1). A4 before step 5: The Asset Coordinator cancels. Nothing is saved or published.
Exception flows	E1 at step 3: A required field is missing or a value is invalid. The system names each problem and returns to step 2. E2 at step 3: The serial or tag number already exists. The system warns, and the Asset Coordinator corrects it or confirms it is a different item. E3 at step 3: A file is not an image or PDF, or is larger than 10 MB. The file is refused and the limits are shown. E4 at step 6: The audit entry cannot be recorded. Nothing is published and an error is shown (FR-7.A6).
Postconditions	Success: The asset has a unique ID and the state Available (or Draft), and the action is in the audit log. Failure: No asset is published. Entered details are kept only if the Asset Coordinator saved a draft.

Use case ID and name	UC-1: Register and Publish an Asset
Rules	R1: Only an Asset Coordinator of the owning department may register its assets. R2: Category must come from the approved list; condition must be New, Good, Fair or Poor. R3: AI suggestions are never saved without the Asset Coordinator’s acceptance.

UC-3: Submit a Resource Request

Use case ID and name	UC-3: Submit a Resource Request
Description	A Department Representative asks for a published asset, or describes a need, for their department. Submitting does not reserve the asset; the request then goes to review (UC-4).
Primary actor (initiator)	Department Representative
Secondary actors	Asset Coordinator of the owning department (notified; reviews the request in UC-4); Procurement Team (finds matches for an unmatched need, UC-9).
Related requirements	FR-2.7; FR-3.1 to FR-3.15; FR-5.9, FR-5.27; FR-7.N1 to FR-7.N6; FR-11.2, FR-11.3; PER-2, PER-3, PER-6; SAF-1, SAF-5; QUA-1, QUA-4, QUA-7; BR-2, BR-5.
Preconditions	The Department Representative is signed in and assigned to a department. For a specific request, the asset is published with an available quantity.
Trigger	The Department Representative chooses Request on a published asset, or creates a request for a need.
Main flow	1. The Department Representative opens the request form; for a specific asset, its details are filled in. 2. The Department Representative enters the category, purpose, quantity, required date, delivery location and any optional details. 3. The Department Representative submits the request. 4. The system checks the values, permissions and the current available quantity. 5. The system gives the request an ID, sets it to Submitted and notifies the owning Asset Coordinator (UC-4). No units are reserved. 6. The Department Representative follows the request state until it is Completed or closed.
Alternative flows	A1 at step 1: No listed asset fits. The request is saved as Open and offered to the Procurement Team for matching (UC-9). A2 after step 5: While the request is Submitted, the Department Representative edits it. Continue at step 4. A3 before handover: The Department Representative cancels. The request becomes Cancelled and any reservation is released. A4 at step 1: The Department Representative prepares the request with the AI assistant (UC-11), checks the final details and confirms. Continue at step 4.
Exception flows	E1 at step 4: A required field is missing or a value is invalid. The system names each problem and returns to step 2. E2 at step 4: The quantity exceeds the current available quantity, or the asset is restricted (under maintenance, pending end-of-life action or unsafe). The system refuses the request and shows the current quantity or restriction; the Department

Use case ID and name	UC-3: Submit a Resource Request
	Representative may revise it. E3 at step 5: Saving fails. No request is created and the system reports the failure. E4 after step 5: The request is not decided within 10 working days. It becomes Expired and both departments are notified.
Postconditions	Success: The request has an ID and the state Submitted (or Open), and the owning Asset Coordinator is notified. The asset's available quantity, ownership and location are unchanged. Failure: No request is created.
Rules	R1: A Department Representative can request only for their own department. R2: The quantity must be a positive whole number no greater than the current available quantity at submission. R3: Submitting a request does not reserve units or change ownership or location. R4: The requesting Department Representative can never release or approve their own request.

UC-5: Transfer an Asset

Use case ID and name	UC-5: Transfer an Asset
Description	After a request is approved, the owning Asset Coordinator hands the asset to the receiving department. The receiver confirms receipt, and custody and location change.
Primary actor (initiator)	Asset Coordinator (of the owning department)
Secondary actors	Department Representative or Asset Coordinator of the receiving department (confirms receipt); Administrator (decides disputes).
Related requirements	FR-1.10; FR-3.9; FR-4.6 to FR-4.16; FR-5.9, FR-5.16, FR-5.27; FR-7.N1 to FR-7.N6; FR-13.3; PER-3, PER-6; SAF-1, SAF-2, SAF-5, SAF-7, SAF-9; SEC-12, SEC-13; QUA-4, QUA-7; BR-5, BR-6.
Preconditions	The request is Approved and the requested quantity is reserved.
Trigger	The Asset Coordinator chooses to arrange the transfer for an approved request.
Main flow	1. The Asset Coordinator enters the method (collection or delivery), planned date, from and to locations, contact person and any transfer cost. 2. The system creates a Scheduled transfer and notifies the receiving department, showing any handling or safety notes. 3. At handover, the Asset Coordinator marks the asset as handed over and attaches any handover documents. 4. The system checks the safety conditions (SAF-5, SAF-9) and sets the transfer and asset to In transfer. 5. A Department Representative or the Asset Coordinator of the receiving department confirms receipt and records the condition on arrival. 6. The system changes the Asset Coordinator, department and location, sets the transfer to Received and the request to Completed, and records each step in the

Use case ID and name	UC-5: Transfer an Asset
	asset history and audit log.
Alternative flows	A1 before step 3: The Asset Coordinator changes the planned date. The receiver is notified. A2 before step 3: The Asset Coordinator or the requesting Department Representative cancels. The transfer becomes Cancelled and the reserved units return to Available. A3 at step 1: Only part of a multi-unit asset is transferred. At step 6, the system creates a linked record for the transferred units (FR-4.15).
Exception flows	E1 at step 5: The asset is damaged or is the wrong item. The receiver reports a problem, the transfer becomes Disputed and the Administrator decides to accept it or return the asset. E2 at step 4: The asset is unsafe, or is a risky asset without a safe inspection in the last 90 days. Handover is blocked (SAF-5, SAF-9). E3 before step 3: No handover within 10 working days of approval. The reservation is released and the request becomes Expired. E4 at step 5: Receipt is not confirmed within 5 working days. The system reminds the receiver and copies the Administrator. E5 at step 5: A user outside the receiving department tries to confirm receipt. The system refuses.
Postconditions	Success: The asset belongs to the receiving department with its new Asset Coordinator and location, and the full transfer is recorded. Failure or cancellation: Custody and location are unchanged, and the asset returns to Available or stays Disputed until decided.
Rules	R1: Only an Approved request can be transferred. R2: Only the receiving department (a Department Representative or its Asset Coordinator) can confirm receipt. R3: Custody and location change only on confirmed receipt.

Appendix B: Analysis Models (State Lists)

Each table lists the states of one record type, what each state means and the states it can move to. The system sets every state; users cannot type a state directly. Final states cannot change, and corrections are recorded as linked records (SEC-12).

B.1 Asset States

State	Meaning	Possible next states
Draft	**Registered but not published; also used while a hazardous asset waits for its safety review**	**Available, Archived**
Available	**Published; some or all units can be requested**	**Reserved, Withdrawn, Under maintenance, Pending Safety Review**
Reserved	**All available units are reserved for approved requests**	**Available, In transfer**
In transfer	**Handed over and not yet received**	**In use, Available (returned after a dispute)**
In use	**Used by its owning department and not listed**	**Available, Under maintenance, Pending Safety Review**
Withdrawn	**Removed from listing by the Asset Coordinator, or because it was marked not safe to use**	**Available, In use, Under maintenance, Pending Safety Review**
Under maintenance	**Has a maintenance request that is Under Inspection or Repair in Progress**	**Available, In use, Withdrawn, Pending Safety Review**
Pending Safety Review	**End-of-life action proposed; waiting for the safety review**	**Pending End-of-Life Approval; previous state if rejected**
Pending End-of-Life Approval	**Safety review recorded; waiting for approval and the completion record**	**Donated, Recycled, Disassembled, Retired, Disposed; previous state if rejected**
Donated	**Given to an organization outside the university (final)**	**None (final state)**
Recycled	**Sent for recycling (final)**	**None (final state)**
Disassembled	**Taken apart; usable components may be registered as new linked assets (final)**	**None (final state)**
Retired	**Taken out of use and kept, for example in central storage (final)**	**None (final state)**
Disposed	**Disposed of as the last option (final)**	**None (final state)**

State	Meaning	Possible next states
Archived	**Draft removed without being published; its ID is never reused (final)**	**None (final state)**

B.2 Request States

State	Meaning	Possible next states
Open	**Need request with no asset chosen; waiting for a match (UC-9)**	**Submitted, Cancelled**
Submitted	**Asset chosen; waiting for the owning Asset Coordinator to release or decline it. No units are reserved**	**Released, Rejected, Cancelled, Expired**
Released	**Released by the Asset Coordinator; waiting for the required approvals**	**Approved, Rejected, Cancelled, Expired**
Approved	**All required approvals recorded; the requested quantity is reserved**	**In transfer, Cancelled, Expired**
In transfer	**Asset handed over; waiting for receipt**	**Completed, Disputed**
Disputed	**The receiver reported a problem; the Administrator decides**	**Completed, Cancelled**
Completed	**Receipt confirmed (final)**	**None (final state)**
Rejected	**Declined by the Asset Coordinator or rejected by an approver, with a reason (final)**	**None (final state)**
Cancelled	**Cancelled before handover by the requesting Department Representative, or with its transfer by the Asset Coordinator, or ended after a dispute (final)**	**None (final state)**
Expired	**Not decided within 10 working days of submission, or not handed over within 10 working days of approval (final)**	**None (final state)**

B.3 Transfer States

State	Meaning	Possible next states
Scheduled	**Method, date and locations agreed**	**In transfer, Cancelled**
In transfer	**Asset handed over**	**Received, Disputed**
Disputed	**Problem reported on arrival; the Administrator decides**	**Received, Returned**

State	Meaning	Possible next states
Received	Receipt confirmed; custody moved (final)	None (final state)
Returned	Asset sent back to the original department (final)	None (final state)
Cancelled	Stopped before handover; reserved units return to Available (final)	None (final state)

B.4 Maintenance States

State	Meaning	Possible next states
Reported	Defect logged (FR-5.1)	Under Inspection, Cancelled
Inspection Requested	Inspection requested (FR-5.2)	Under Inspection, Cancelled
Under Inspection	The assigned Maintenance Staff member has started the inspection	Repair in Progress, Completed, Closed – Not Repairable, Cancelled
Repair in Progress	Repair or refurbishment under way	Completed, Closed – Not Repairable, Cancelled
Completed	Work finished; updated condition recorded and asset returned to service (final)	None (final state)
Closed – Not Repairable	Cannot be repaired; asset flagged for end-of-life review (final)	None (final state)
Cancelled	Cancelled with a reason before completion (final)	None (final state)

B.5 End-of-Life Proposal States

State	Meaning	Possible next states
Pending Safety Review	Proposal submitted; waiting for a Sustainability Officer other than the proposer (FR-5.23)	Pending End-of-Life Approval, Rejected
Pending End-of-Life Approval	Safety review recorded; waiting for Administrator approval (BR-8)	Approved, Rejected
Approved	All approvals recorded; waiting for the completion record	Completed
Completed	Completion record saved; asset in its final state (final)	None (final state)

State	Meaning	Possible next states
Rejected	**Rejected at safety review or approval, with a reason; asset returns to its previous state (final)**	**None (final state)**

B.6 Account States

State	Meaning	Possible next states
Pending approval	**Application with recorded consent; waiting for activation (UC-13)**	**Active, Closed**
Active	**Account can sign in and use its permitted functions**	**Suspended, Closed**
Suspended	**Access stopped temporarily by a System Administrator**	**Active, Closed**
Closed	**Access ended, for example after consent is withdrawn; history kept (final)**	**None (final state)**

B.7 AI Report States

State	Meaning	Possible next states
Draft	**AI-generated draft waiting for review (FR-12.9)**	**Published, Discarded**
Published	**Approved and shared with permitted recipients (final)**	**None (final state)**
Discarded	**Rejected by the reviewer and kept for reference (final)**	**None (final state)**

Appendix D: Requirements Traceability

Table D-1 traces every requirement ID in this document to its Phase 2 source [5], the main stakeholder role, the linked use cases and its priority. Consecutive requirements that share all four values are grouped on one row. Table D-2 shows how the requirements respond to each risk in the Phase 2 risk register.

Table D-1: Requirements traceability matrix

Requirement ID	Phase 2 source	Main stakeholder	Use case	Priority
FR-1.1 to FR-1.8	FR-1	Asset Coordinator	UC-1	Essential
FR-1.9	FR-1	Asset Coordinator	UC-1, UC-2	Essential
FR-1.10	FR-1	Asset Coordinator	UC-1, UC-5	Essential
FR-1.11	FR-1	Asset Coordinator	UC-1	Desirable
FR-1.12	FR-1	Asset Coordinator	UC-1, UC-10	Essential
FR-1.13	FR-1	Asset Coordinator	UC-1	Essential
FR-1.14	FR-1	Asset Coordinator	UC-1	Future
FR-1.15	FR-1	Asset Coordinator	UC-1	Essential
FR-2.1	FR-2	Department Representative, Asset Coordinator	UC-1, UC-2	Essential
FR-2.2 to FR-2.6	FR-2	Department Representative, Asset Coordinator	UC-2	Essential
FR-2.7	FR-2	Department Representative, Asset Coordinator	UC-2, UC-3	Essential
FR-2.8	FR-2	Department Representative, Asset Coordinator	UC-2	Desirable
FR-3.1	FR-3	Department Representative	UC-3	Essential
FR-3.2	FR-3	Department Representative	UC-3, UC-9	Essential
FR-3.3 to FR-3.7	FR-3	Department Representative	UC-3	Essential
FR-3.8	FR-3	Department Representative	UC-3, UC-4	Essential
FR-3.9	FR-3	Department Representative	UC-3, UC-4, UC-5	Desirable
FR-3.10 to FR-3.11	FR-3	Department Representative	UC-3	Essential
FR-3.12	FR-3	Department Representative	UC-3, UC-4	Essential

Requirement ID	Phase 2 source	Main stakeholder	Use case	Priority
FR-3.13	FR-3	Department Representative	UC-3	Desirable
FR-3.14	FR-3	Department Representative	UC-3, UC-11	Desirable
FR-3.15	FR-3	Department Representative	UC-3	Essential
FR-4.1 to FR-4.2	FR-4	Asset Coordinator, Administrator, Procurement Team	UC-4	Essential
FR-4.3	FR-4	Asset Coordinator, Administrator, Procurement Team	UC-4	Desirable
FR-4.4 to FR-4.5	FR-4	Asset Coordinator, Administrator, Procurement Team	UC-4	Essential
FR-4.6	FR-4	Asset Coordinator, Administrator, Procurement Team	UC-4, UC-5	Essential
FR-4.7	FR-4	Asset Coordinator, Administrator, Procurement Team	UC-5	Essential
FR-4.8	FR-4	Asset Coordinator, Administrator, Procurement Team	UC-5	Desirable
FR-4.9 to FR-4.12	FR-4	Asset Coordinator, Administrator, Procurement Team	UC-5	Essential
FR-4.13	FR-4	Asset Coordinator, Administrator, Procurement Team	UC-5	Desirable
FR-4.14	FR-4	Asset Coordinator, Administrator, Procurement Team	UC-4, UC-5	Essential
FR-4.15 to FR-4.16	FR-4	Asset Coordinator, Administrator, Procurement Team	UC-5	Essential
FR-5.1 to FR-5.8	FR-5	Maintenance Staff, Asset Coordinator, Sustainability Officer, Administrator	UC-6	Essential
FR-5.9	FR-5	Maintenance Staff, Asset Coordinator, Sustainability Officer, Administrator	UC-2, UC-3, UC-5, UC-6, UC-9	Essential

Requirement ID	Phase 2 source	Main stakeholder	Use case	Priority
FR-5.10	FR-5	Maintenance Staff, Asset Coordinator, Sustainability Officer, Administrator	UC-6	Essential
FR-5.11	FR-5	Maintenance Staff, Asset Coordinator, Sustainability Officer, Administrator	UC-6, UC-7	Desirable
FR-5.12	FR-5	Maintenance Staff, Asset Coordinator, Sustainability Officer, Administrator	UC-6, UC-7	Essential
FR-5.13 to FR-5.15	FR-5	Maintenance Staff, Asset Coordinator, Sustainability Officer, Administrator	UC-6	Essential
FR-5.16	FR-5	Maintenance Staff, Asset Coordinator, Sustainability Officer, Administrator	UC-1, UC-4, UC-5, UC-6, UC-7	Essential
FR-5.17 to FR-5.18	FR-5	Maintenance Staff, Asset Coordinator, Sustainability Officer, Administrator	UC-6, UC-7	Essential
FR-5.19	FR-5	Maintenance Staff, Asset Coordinator, Sustainability Officer, Administrator	UC-2, UC-7	Desirable
FR-5.20 to FR-5.21	FR-5	Maintenance Staff, Asset Coordinator, Sustainability Officer, Administrator	UC-7	Essential
FR-5.22	FR-5	Maintenance Staff, Asset Coordinator, Sustainability Officer, Administrator	UC-7	Desirable
FR-5.23 to FR-5.26	FR-5	Maintenance Staff, Asset Coordinator, Sustainability Officer, Administrator	UC-7	Essential
FR-5.27	FR-5	Maintenance Staff, Asset Coordinator, Sustainability Officer, Administrator	UC-2, UC-3, UC-5, UC-7, UC-9	Essential
FR-5.28 to FR-5.32	FR-5	Maintenance Staff, Asset Coordinator, Sustainability Officer, Administrator	UC-7	Essential
FR-5.33	FR-5	Maintenance Staff, Asset Coordinator, Sustainability Officer, Administrator	UC-1, UC-7	Future

Requirement ID	Phase 2 source	Main stakeholder	Use case	Priority
FR-5.34	FR-5	Maintenance Staff, Asset Coordinator, Sustainability Officer, Administrator	UC-7	Essential
FR-6.1 to FR-6.2	FR-6	System Administrator, Applicant	UC-13	Essential
FR-6.3 to FR-6.4	FR-6	System Administrator, Applicant	UC-8, UC-13	Essential
FR-6.5	FR-6	System Administrator, Applicant	UC-1 to UC-13	Essential
FR-6.6	FR-6	System Administrator, Applicant	UC-13	Essential
FR-6.7	FR-6	System Administrator, Applicant	UC-1 to UC-13	Essential
FR-6.8 to FR-6.12	FR-6	System Administrator, Applicant	UC-8	Essential
FR-7.N1 to FR-7.N5	FR-7	All roles	UC-3 to UC-9, UC-12, UC-13	Essential
FR-7.N6	FR-7	All roles	UC-3, UC-4, UC-5	Desirable
FR-7.R1 to FR-7.R6	FR-7	Report users (Section 5.1)	UC-12	Essential
FR-7.R7	FR-7	Report users (Section 5.1)	UC-12	Desirable
FR-7.R8	FR-7	Report users (Section 5.1)	UC-12	Future
FR-7.A1 to FR-7.A2	FR-7	System Administrator, Administrator	UC-1 to UC-13	Essential
FR-7.A3	FR-7	System Administrator, Administrator	UC-8	Essential
FR-7.A4	FR-7	System Administrator, Administrator	UC-7, UC-9, UC-10, UC-11, UC-12	Essential
FR-7.A5 to FR-7.A6	FR-7	System Administrator, Administrator	UC-1 to UC-13	Essential
FR-8.1 to FR-8.9	FR-8	Procurement Team	UC-9	Essential
FR-9.1 to FR-9.9	FR-9	Asset Coordinator	UC-10	Essential

Requirement ID	Phase 2 source	Main stakeholder	Use case	Priority
FR-10.1 to FR-10.8	FR-10	Sustainability Officer, Maintenance Staff	UC-7	Desirable
FR-11.1	FR-11	Department Representative	UC-11	Desirable
FR-11.2 to FR-11.3	FR-11	Department Representative	UC-3, UC-11	Desirable
FR-11.4 to FR-11.10	FR-11	Department Representative	UC-11	Desirable
FR-12.1	FR-12	Finance Officer, Sustainability Officer	UC-12	Essential
FR-12.2	FR-12	Finance Officer, Sustainability Officer	UC-12	Essential (a, b); Desirable (c–g)
FR-12.3 to FR-12.7	FR-12	Finance Officer, Sustainability Officer	UC-12	Essential
FR-12.8	FR-12	Finance Officer, Sustainability Officer	UC-12	Desirable
FR-12.9 to FR-12.12	FR-12	Finance Officer, Sustainability Officer	UC-12	Essential
FR-13.1 to FR-13.2	New in Phase 3	Finance Officer, Sustainability Officer, University Management	UC-12	Essential
FR-13.3	New in Phase 3	Finance Officer, Sustainability Officer, University Management	UC-5, UC-6, UC-7, UC-12	Essential
FR-13.4 to FR-13.5	New in Phase 3	Finance Officer, Sustainability Officer, University Management	UC-12	Essential
FR-13.6	New in Phase 3	Finance Officer, Sustainability Officer, University Management	UC-1, UC-7, UC-12	Essential
FR-13.7 to FR-13.12	New in Phase 3	Finance Officer, Sustainability Officer, University Management	UC-12	Essential
FR-13.13	New in Phase 3	Finance Officer, Sustainability Officer, University Management	UC-12	Desirable

Requirement ID	Phase 2 source	Main stakeholder	Use case	Priority
AI-1 to AI-7	FR-8 to FR-12	All users of AI features	UC-7, UC-9 to UC-12	Essential
AI-8	FR-8 to FR-12	All users of AI features	UC-7, UC-9, UC-11, UC-12	Essential
AI-9 to AI-10	FR-8 to FR-12	All users of AI features	UC-7, UC-9 to UC-12	Essential
PER-1	Phase 2 technical feasibility	All users; IT Department	UC-1 to UC-13	Essential
PER-2	Phase 2 technical feasibility	All users; IT Department	UC-2, UC-3, UC-11	Essential
PER-3	Phase 2 technical feasibility	All users; IT Department	UC-1, UC-3 to UC-8, UC-13	Essential
PER-4 to PER-5	Phase 2 technical feasibility	All users; IT Department	UC-7, UC-9 to UC-12	Essential
PER-6	Phase 2 technical feasibility	All users; IT Department	UC-1 to UC-7, UC-9 to UC-11	Essential
SAF-1	Phase 2 risks and ethics	Sustainability Officer, Maintenance Staff, Facilities Management	UC-1, UC-2, UC-3, UC-5, UC-7, UC-9	Essential
SAF-2	Phase 2 risks and ethics	Sustainability Officer, Maintenance Staff, Facilities Management	UC-4 to UC-7, UC-9 to UC-12	Essential
SAF-3 to SAF-4	Phase 2 risks and ethics	Sustainability Officer, Maintenance Staff, Facilities Management	UC-7	Essential
SAF-5	Phase 2 risks and ethics	Sustainability Officer, Maintenance Staff, Facilities Management	UC-1, UC-2, UC-3, UC-5, UC-6	Essential
SAF-6	Phase 2 risks and ethics	Sustainability Officer, Maintenance Staff, Facilities Management	UC-7	Essential
SAF-7	Phase 2 risks and ethics	Sustainability Officer, Maintenance Staff, Facilities Management	UC-5, UC-7	Desirable

Requirement ID	Phase 2 source	Main stakeholder	Use case	Priority
SAF-8	Phase 2 risks and ethics	Sustainability Officer, Maintenance Staff, Facilities Management	UC-1 to UC-13	Essential
SAF-9	Phase 2 risks and ethics	Sustainability Officer, Maintenance Staff, Facilities Management	UC-5, UC-6	Essential
SEC-1	FR-6; Phase 2 legal analysis	System Administrator; IT Department	UC-1 to UC-13	Essential
SEC-2	FR-6; Phase 2 legal analysis	System Administrator; IT Department	UC-13	Essential
SEC-3 to SEC-4	FR-6; Phase 2 legal analysis	System Administrator; IT Department	UC-1 to UC-13	Essential
SEC-5	FR-6; Phase 2 legal analysis	System Administrator; IT Department	UC-2, UC-6, UC-7, UC-8, UC-9, UC-12	Essential
SEC-6 to SEC-8	FR-6; Phase 2 legal analysis	System Administrator; IT Department	UC-8, UC-13	Essential
SEC-9	FR-6; Phase 2 legal analysis	System Administrator; IT Department	UC-8	Essential
SEC-10	FR-6; Phase 2 legal analysis	System Administrator; IT Department	UC-7, UC-9	Essential
SEC-11	FR-6; Phase 2 legal analysis	System Administrator; IT Department	UC-9, UC-10, UC-11, UC-12	Essential
SEC-12	FR-6; Phase 2 legal analysis	System Administrator; IT Department	UC-4, UC-5, UC-6, UC-7	Essential
SEC-13	FR-6; Phase 2 legal analysis	System Administrator; IT Department	UC-1, UC-5, UC-6, UC-10	Essential
QUA-1	Phase 2 technical feasibility	All users; IT Department	UC-1, UC-3, UC-9, UC-10, UC-11	Essential
QUA-2 to QUA-3	Phase 2 technical feasibility	All users; IT Department	UC-1 to UC-13	Essential
QUA-4	Phase 2 technical feasibility	All users; IT Department	UC-1, UC-3 to UC-7, UC-9 to UC-11	Essential
QUA-5 to QUA-6	Phase 2 technical feasibility	All users; IT Department	UC-1 to UC-13	Essential

Requirement ID	Phase 2 source	Main stakeholder	Use case	Priority
QUA-7	Phase 2 technical feasibility	All users; IT Department	UC-1 to UC-7, UC-9, UC-10	Essential
QUA-8	Phase 2 technical feasibility	All users; IT Department	UC-1 to UC-13	Essential
QUA-9	Phase 2 technical feasibility	All users; IT Department	UC-10	Essential
QUA-10	Phase 2 technical feasibility	All users; IT Department	UC-9	Essential
QUA-11	Phase 2 technical feasibility	All users; IT Department	UC-7	Desirable
QUA-12 to QUA-13	Phase 2 technical feasibility	All users; IT Department	UC-7, UC-9	Essential
QUA-14	Phase 2 technical feasibility	All users; IT Department	UC-7, UC-9 to UC-12	Essential
QUA-15	Phase 2 technical feasibility	All users; IT Department	UC-11	Desirable
BR-1	Phase 2 stakeholder roles	Head of Facilities Management; approving roles	None (policy rule)	Essential
BR-2	Phase 2 stakeholder roles	Head of Facilities Management; approving roles	UC-1 to UC-4	Essential
BR-3	Phase 2 stakeholder roles	Head of Facilities Management; approving roles	UC-4	Essential
BR-4	Phase 2 stakeholder roles	Head of Facilities Management; approving roles	UC-4, UC-7, UC-8	Essential
BR-5	Phase 2 stakeholder roles	Head of Facilities Management; approving roles	UC-3 to UC-5	Essential
BR-6	Phase 2 stakeholder roles	Head of Facilities Management; approving roles	UC-5	Essential
BR-7	Phase 2 stakeholder roles	Head of Facilities Management; approving roles	UC-6	Essential

Requirement ID	Phase 2 source	Main stakeholder	Use case	Priority
BR-8	Phase 2 stakeholder roles	Head of Facilities Management; approving roles	UC-7	Essential
BR-9	Phase 2 stakeholder roles	Head of Facilities Management; approving roles	UC-7, UC-9 to UC-12	Essential
BR-10	Phase 2 stakeholder roles	Head of Facilities Management; approving roles	UC-9, UC-12	Essential
BR-11	Phase 2 stakeholder roles	Head of Facilities Management; approving roles	UC-12	Essential
BR-12	Phase 2 stakeholder roles	Head of Facilities Management; approving roles	UC-1 to UC-13	Essential
LEG-1	Phase 2 legal analysis	University legal contact; IT Department	UC-8, UC-9, UC-11, UC-13	Essential
LEG-2	Phase 2 legal analysis	University legal contact; IT Department	None (process requirement)	Essential
LEG-3	Phase 2 legal analysis	University legal contact; IT Department	UC-9, UC-10, UC-11, UC-12	Essential
LEG-4	Phase 2 legal analysis	University legal contact; IT Department	UC-1 to UC-13	Essential
LEG-5	Phase 2 legal analysis	University legal contact; IT Department	UC-12	Essential
LEG-6	Phase 2 legal analysis	University legal contact; IT Department	UC-9, UC-12	Essential
LEG-7	Phase 2 legal analysis	University legal contact; IT Department	UC-7	Essential
LEG-8	Phase 2 legal analysis	University legal contact; IT Department	UC-12	Desirable
LEG-9	Phase 2 legal analysis	University legal contact; IT Department	UC-9	Essential
ORG-1	Phase 2 operational feasibility	IT Department; development team	None (process requirement)	Future (required before a real pilot)
ORG-2	Phase 2 operational feasibility	IT Department; development team	None (process requirement)	Essential

Requirement ID	Phase 2 source	Main stakeholder	Use case	Priority
ORG-3	Phase 2 operational feasibility	IT Department; development team	UC-8, UC-13	Essential
ORG-4	Phase 2 operational feasibility	IT Department; development team	None (process requirement)	Essential

Table D-2: Phase 2 risks and the requirements that address them

Phase 2 risk [5]	Requirements that address it
R1: Insufficient or poor-quality data for AI functions	FR-1.2, FR-1.5 (complete, validated records); FR-9.4 (missing details flagged); FR-13.8 (missing data excluded and listed); ORG-2 (synthetic or reviewed test data); QUA-9, QUA-10 (separate labeled test sets).
R2: AI classification or matching gives inaccurate results	AI-1, AI-10 (confidence shown); AI-3, FR-8.7, FR-9.6 (human review and override); AI-6, FR-8.9 (standard search fallback); QUA-9, QUA-10 (accuracy targets).
R3: AI or LLM service fails or is unavailable	AI-6; PER-5, PER-6; FR-9.8, FR-11.9, FR-12.11 (a manual or standard option keeps the core prototype usable).
R4: Project scope is too large for the semester	Priorities in every requirement table (1.2): Essential requirements form the first prototype; FR-10, FR-11 and the Desirable requirements of other features can be deferred; FR-1.14, FR-5.33, FR-7.R8 and ORG-1 are Future.
R5: Integration problems between modules	ORG-4 (shared repository and regular commits); QUA-4 (records stay correct under retries and simultaneous updates); QUA-7 (category and rule changes without workflow changes).
R6: Unauthorized access or exposure of information	FR-6.5, FR-6.7; SEC-1 to SEC-9, SEC-11; FR-7.A1 to FR-7.A6; LEG-1, LEG-4.
R7: Bias or unfair prioritization	FR-8.5, LEG-9, QUA-13 (department name and size are not ranking factors); AI-3, FR-8.7 (authorized override); BR-9.
R8: Recommendations cannot be explained	AI-2, FR-8.4 (reasons for every match); AI-1 (confidence); QUA-14 (usefulness of explanations); SEC-10 (human review on request).
R9: Web or application security vulnerabilities	SEC-2 to SEC-4, SEC-6; SEC-13 (upload limits, input handling and OWASP test); FR-7.A6.
R10: Prompt injection or disclosure through the assistant	AI-7, AI-9; FR-11.3, FR-11.8 (confirmation and permission checks before any action); SEC-11.
R11: Team availability or uneven workload	A project-management risk handled in the Phase 1 plan [4]; ORG-4 makes each member’s contribution visible.

Phase 2 risk [5]	Requirements that address it
R12: Unexpected infrastructure or AI-service cost	A financial risk handled in the Phase 2 feasibility study [5]; TBD-9 keeps the choice of AI services under IT approval.
R13: Misleading sustainability or carbon estimates	FR-13.2, FR-13.7, FR-13.8, FR-13.10, FR-13.12; FR-12.4, FR-12.5; AI-8; TBD-8.
R14: Users find the workflows difficult to adopt	QUA-1 to QUA-3 (usability and accessibility targets); FR-2.6, FR-3.4 (clear validation messages); TBD-15 (confirmation with real stakeholders).

AI Contributions

Mohammad Alsuwaidi

An AI assistant was used to help draft and organize FR-1, FR-3, FR-4, the data requirements (4.1), the state lists (Appendix B), the traceability matrix (Appendix D) and UC-1, UC-3 and UC-5, and to check them against the project description and the Phase 2 report. All content was reviewed and edited by the author.